# URBAN AIR-QUALITY CONSISTENCY PROFILING

## with Relaxed Functional Dependencies

A reproducible analytical workflow for approximate environmental rules

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KDIID · Final project presentation

**Consistency—not prediction, not causality**

# Why deterministic rules are the wrong target

## Urban environmental data are:

- noisy and affected by measurement uncertainty
- variable across time, sites, and weather conditions
- influenced by factors that are not fully observed

## The analytical question

**When observations are similar in context and pollutant levels, how often do other measurements remain similar?**

Measure approximate regularity—without forcing exact laws.

**The goal is to quantify approximate consistency, not to predict outcomes or prove causality.**

# A conceptual Urban Digital Twin

The Digital Twin is a framing device for an analytical loop:

01
**HISTORICAL DATA**
Two stations, pollutants, meteorology

02
**APPROXIMATE RULES**
RFDs describe expected similarity

03
**VIOLATIONS**
Cases that break the expected pattern

04
**MONITORING**
Monthly metrics reveal regime changes

**Not built here** 3D platform · real-time infrastructure · operational control system

**The contribution is a reproducible analytical workflow—not a 3D or real-time platform.**

# Dataset design: full profiling, DiMε relation

## Beijing Multi-Site Air Quality Data Set

Aotizhongxin + Changping · March 2013–February 2017

| 66,619 | 11 | 1,676 | 1,403,650 |
| --- | --- | --- | --- |
| clean observations | final variables | weekly profiles | pairwise comparisons |

**Missing-value policy** summarize missingness → drop incomplete rows on selected columns → no imputation

**All 66,619 rows feed weekly station × time-slot medians; DiMε uses eight projected attributes.**

# From exact equality to expected similarity

## CLASSICAL FD

$$X_1 = X_2$$

implies

$$Y_1 = Y_2$$

exact constraints

## RELAXED FD

$$X_1 \approx X_2$$

implies expected similarity

$$Y_1 \approx Y_2$$

threshold-based regularities

**RFDs match noisy environmental observations better than deterministic functional dependencies.**

# How similarity is defined

### CATEGORICAL ATTRIBUTES

station, time_slot → exact match

### EXTENT

g3 ≤ 0.10 · greedy mode

### NUMERICAL ATTRIBUTES

absolute distance thresholds

**MEDIUM SETUP · MAIN CONFIGURATION**

| | | | |
|---|---|---|---|
| TEMP | ≤ 2.0 | NO2 | ≤ 10.0 |
| WSPM | ≤ 1.0 | O3 | ≤ 10.0 |
| PM2.5 | ≤ 10.0 | g3 | ≤ 0.10 |
| PM10 | ≤ 15.0 | | |

Medium similarity thresholds are DiMε inputs.

> **Thresholds are supplied to DiMε; inferring thresholds would instead be DOMINO's role.**

# Course algorithm used: DiMε

Full course algorithm—no LHS cap and no custom discovery shortcut.

01
**DIFFERENCE MATRICES**
Absolute distance + categorical equality

02
**SIMILAR SUBSETS**
Pattern subsets; singleton stripping

03
**COMPLETE LATTICE**
C+ candidates, joins, pruning

04
**g3 VALIDATION**
Greedy vertex cover at ε = 0.10

**Inputs** 8 attributes · similarity thresholds · extent threshold ε

**Outputs: minimal RFDs + g3. Iris check: official JAR and Python match exactly, 5/5.**

# Post-discovery metrics keep interpretation honest

### SUPPORT

**antecedent pairs / all pairs**

How often are tuples comparable?

### CONFIDENCE

**valid pairs / antecedent pairs**

How often does the rule hold?

### BASELINE

**mean confidence over 30 seeded RHS permutations**

What happens by chance?

**LIFT**

$$\text{LIFT} = \frac{\text{confidence}}{\text{baseline confidence}}$$

**> 1 means more informative than chance**

**DiMε discovers with g3; support, confidence, baseline, and lift rank its outputs afterward.**

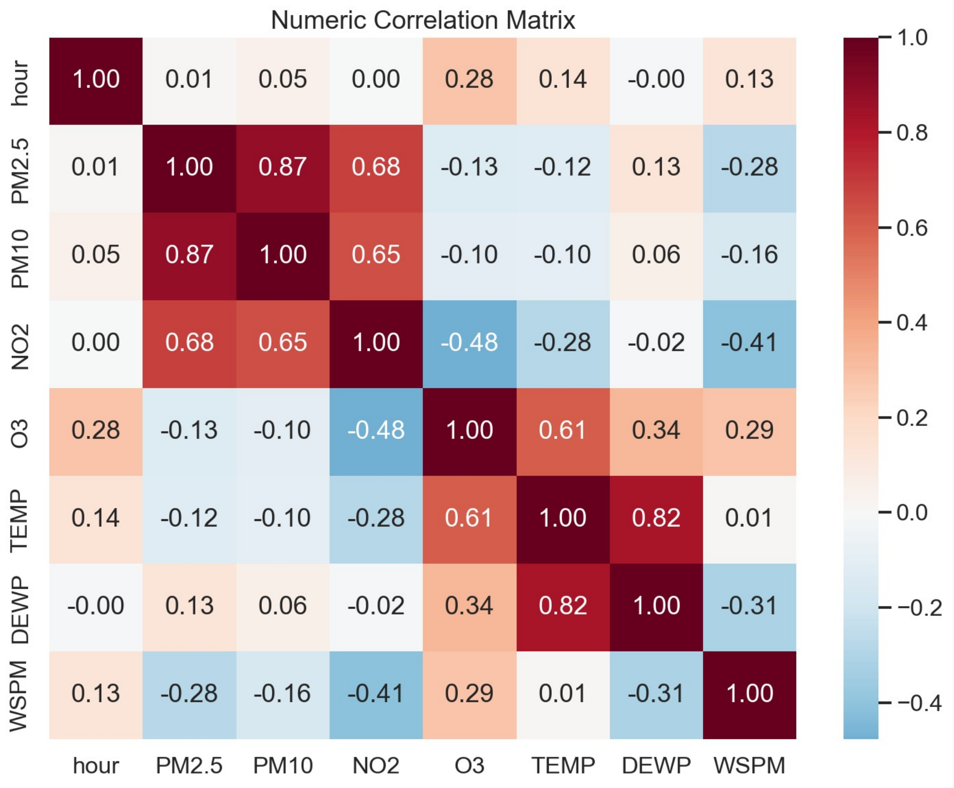
# DiMε discovery is the central pipeline step

01 Preprocess → 02 Profile → 03 Weekly projection → 04 DiMε discovery → 05 Permutation baseline → 06 Bootstrap → 07 Temporal + monthly → 08 Violations

PREPARE | DISCOVER | VALIDATE | INVESTIGATE

**Discovery is DiMε; every baseline and robustness check is strictly post-discovery.**

# Descriptive profiling reveals one dominant pattern



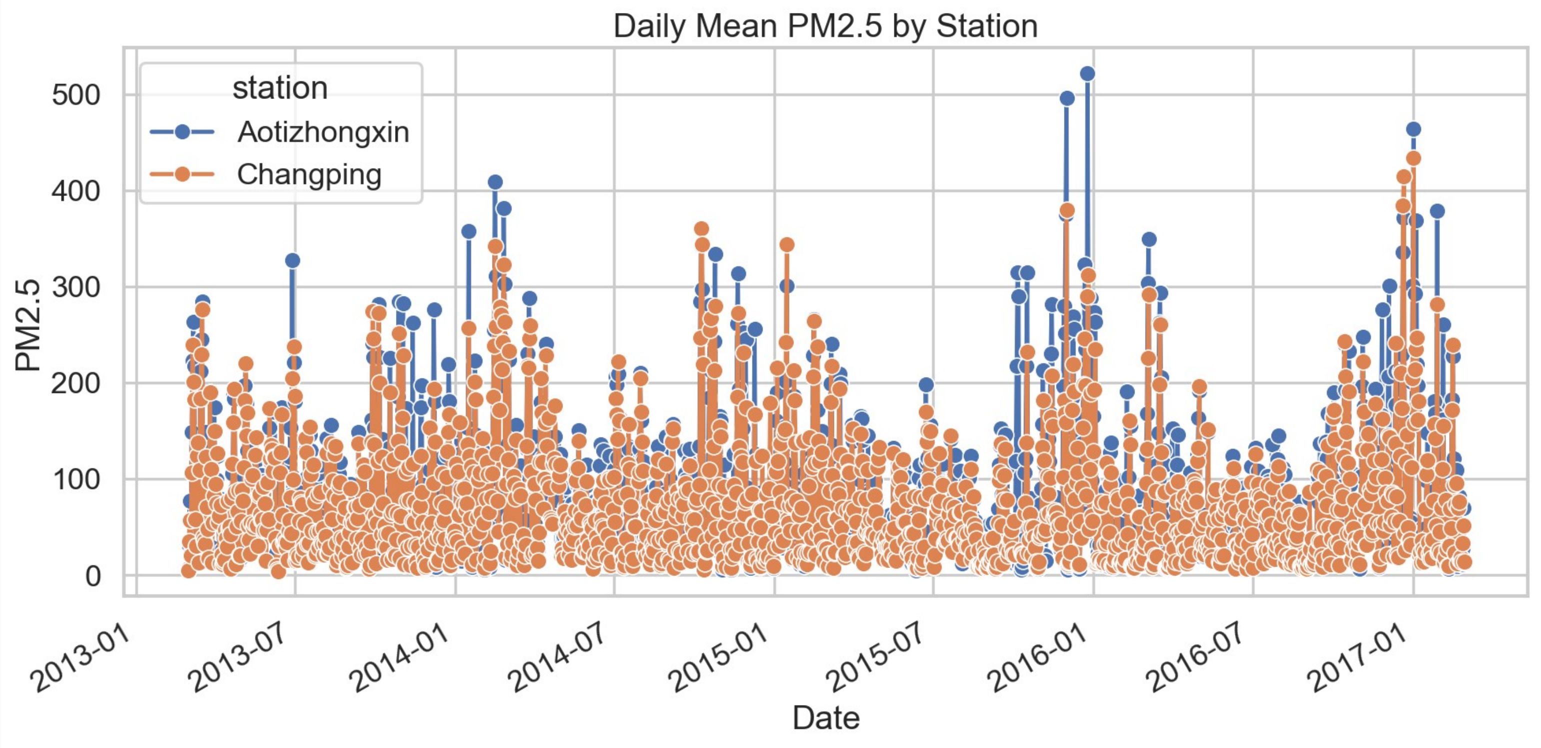

## 0.874

**PM2.5–PM10**
**Pearson correlation**

Correlation is descriptive context—not an RFD metric and not causal evidence.

**Fine and coarse particulate matter form the clearest descriptive relationship in the dataset.**

# PM2.5 behavior changes across time and stations




**Temporal and site-level variability motivates continuous profiling rather than one static summary.**

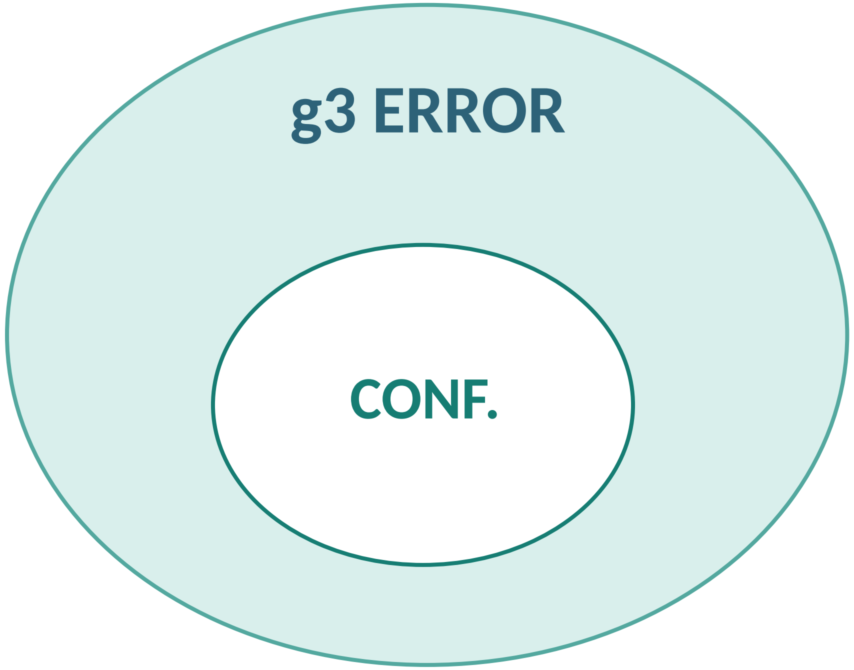
# Top RFDs discovered by DiMε

| Rule | Support | Conf. | g3 | Lift |
|---|---|---|---|---|
| **station, PM10, NO2, O3, TEMP → WSPM** | **0.0016** | **0.899** | **0.092** | **1.163** |
| PM2.5, PM10, NO2, O3, TEMP → WSPM | 0.0013 | 0.889 | 0.074 | **1.155** |
| time_slot → WSPM | 0.2496 | 0.858 | 0.047 | **1.113** |

**5 minimal RFDs · 960 candidates · 8 lattice levels**

**Four of five RFDs have low support; time_slot → WSPM is most general (support 0.24955).**

# g3 and confidence answer different questions




**Minimum tuple-removal fraction needed to make an RFD hold.**

- Valid similar pairs divided by antecedent pairs.
- DiMε discovers rules with g3 ≤ 0.10.
- Pairwise confidence ranks discovered rules afterward.

**DISCOVERY ≠ RANKING**

**Low g3 can coexist with pair violations because tuple removal and pair counting differ.**

# Manual candidates are supplementary

station, PM2.5, NO2 → PM10

| 0.025 | 0.483 | 0.333 | 3.24 |
|---|---|---|---|
| support | confidence | baseline | lift |

This rule was specified manually and validated on the legacy sample.

**It is not an output of DiMε and does not drive the main results.**

> **Manual and binned variants remain only as transparent supplementary comparisons.**

# The top DiMε rule remains relatively stable

## BOOTSTRAP · 30 BALANCED RESAMPLES

**0.930**
mean confidence

**1.207**
mean lift

95% empirical lift interval

**[1.171, 1.244]**

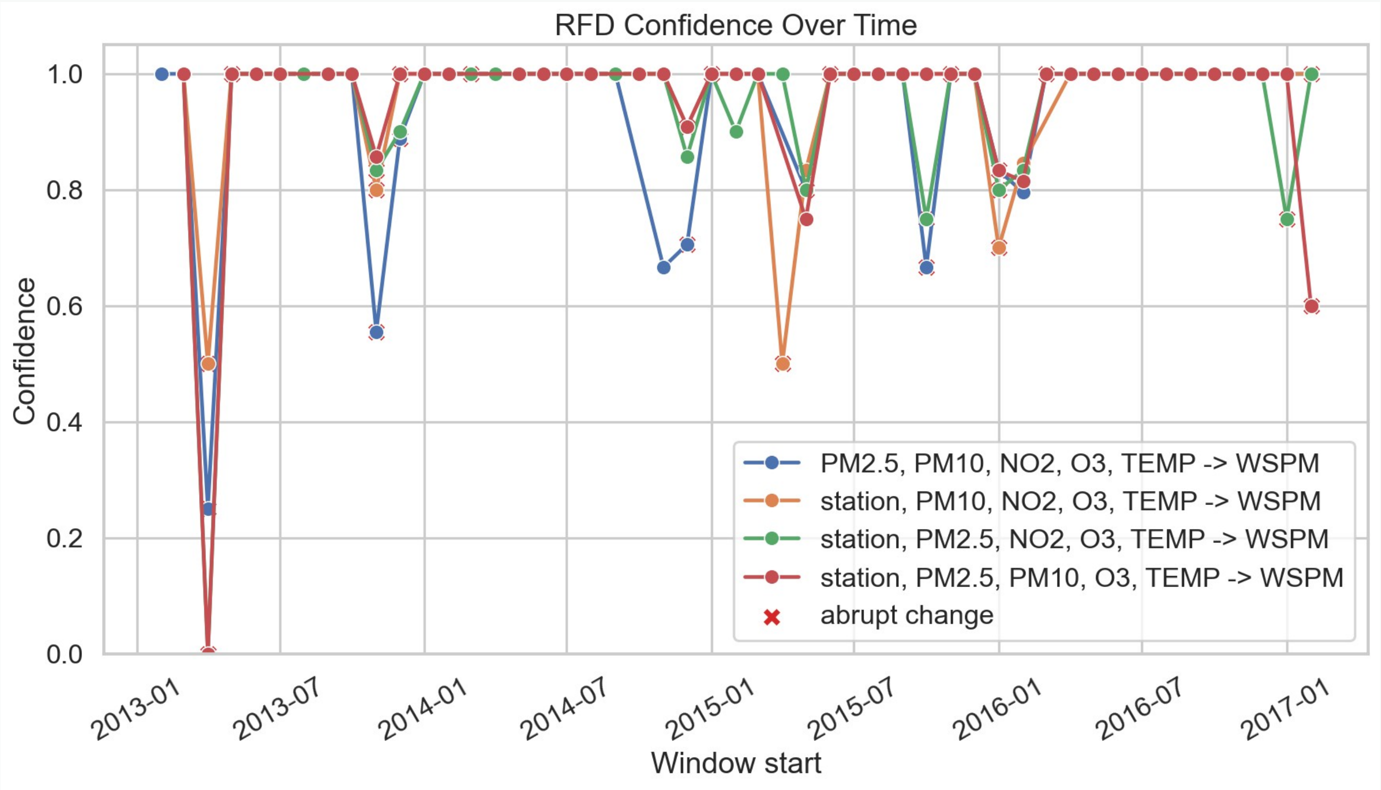
## CHRONOLOGICAL VALIDATION · 75 / 25

| TRAIN | | TEST |
|---|---|---|
| Mar 2013–Feb 2016 | | Mar 2016–Feb 2017 |
| **lift 1.161** | → | **lift 1.178** |

Confidence rises from 0.888 to 0.937.

**Bootstrap and the held-out year keep lift above 1 for the top DiMε rule.**

# Continuous profiling exposes changing regimes




**49**

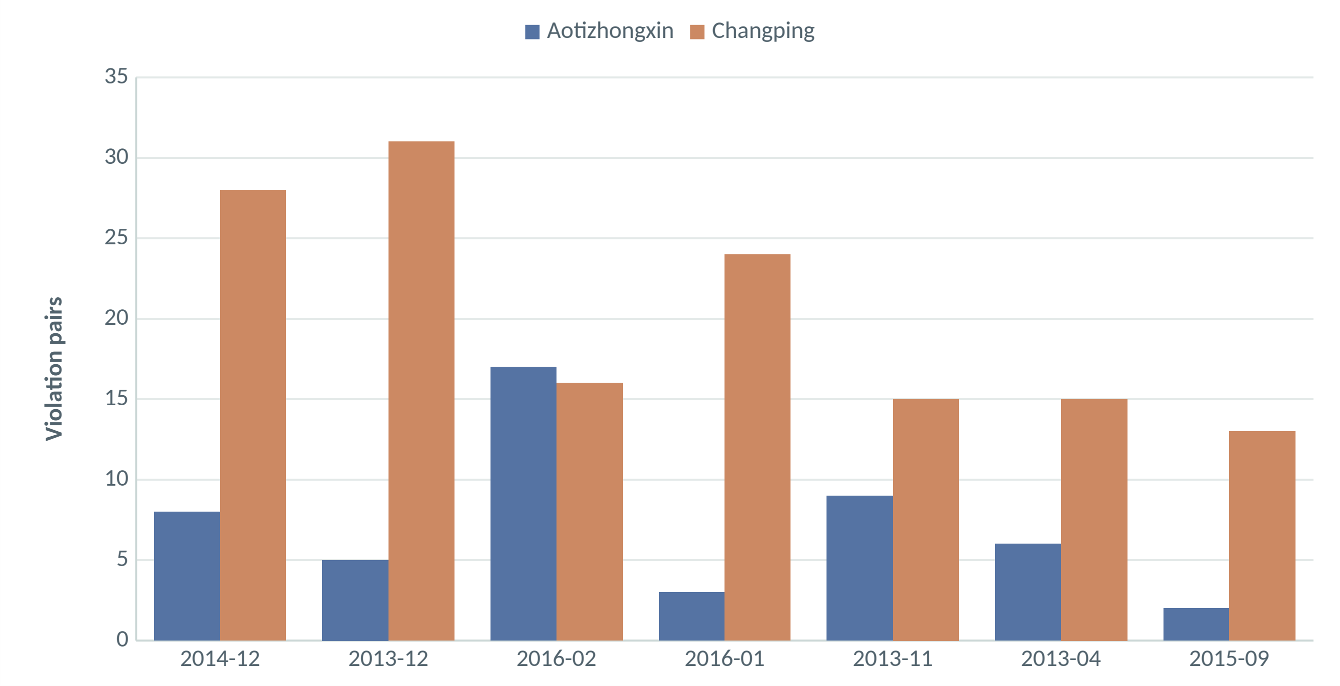
**monthly windows**

- confidence varies over time
- support changes with comparability
- pair counts affect reliability

**Monthly metrics are recalculated only for RFDs discovered by DiMε.**

# Violations are investigation targets—not just errors

Top 2 DiMε RFDs · 200 strongest pairs each




## A violation may reflect:

- local site heterogeneity
- unobserved local conditions
- meteorology outside the rule
- weekly aggregation effects
- measurement or data-quality issues

Top group: Changping · Jan 2016 · afternoon · 9 pairs.

**The value of violations lies in targeted follow-up, not automatic labeling.**

# What DiMε contributes

- Full course algorithm: difference matrices, lattice, C+, g3, pruning.
- **5 minimal RFDs from 960 candidates across all 8 lattice levels.**
- Baseline, lift, bootstrap and temporal checks are post-discovery.
- The weekly projection uses every one of the 66,619 cleaned rows.
- **Four of five RFDs have low support and describe specific regularities.**

> **“The contribution is exact algorithm fidelity: DiMε discovers the rules; every other metric explains their scope and stability.”**

Thank you

# Backup · Sources

01 **Beijing Multi-Site Air Quality Data Set, UCI Machine Learning Repository**

02 KDIID Lecture 3, Relaxed Functional Dependencies · DiMε slides 29–41

03 Caruccio, Deufemia, Polese, “Mining relaxed functional dependencies from data”, Data Mining and Knowledge Discovery, 2020